



Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to drive strategic decisions



OVERVIEW

Project Scope

Our Mission

Analyze transactional data to uncover spending patterns, customer segments, and product preferences

3.9K

Purchases

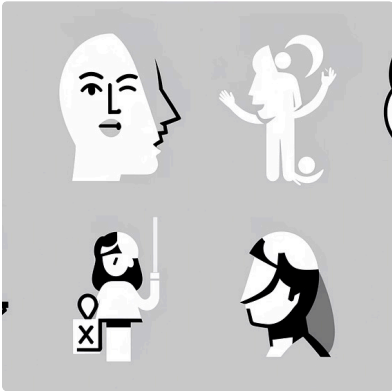
18

Data Points

4

Categories

Dataset at a Glance



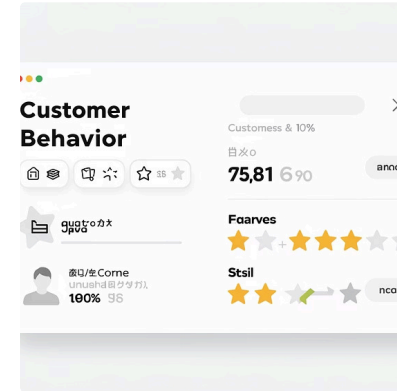
Demographics

Age, gender, location, subscription status



Purchase Details

Item, category, amount, season, size, color



Behavior Patterns

Discounts, reviews, frequency, shipping preferences

❏ Missing data: 37 values in Review Rating column (handled via median imputation)

Data Preparation Process



Systematic approach to transform raw data into analysis-ready format

Cleaning

Standardized columns, imputed missing ratings by category median

Engineering

Created age groups and purchase frequency metrics

Integration

Connected to PostgreSQL for advanced analysis



 SQL ANALYSIS

Key Business Insights

01

Revenue by Gender

Male customers: \$157,890 | Female customers: \$75,191

03

Top Products

Gloves (3.86), Sandals (3.84), Boots (3.82) lead ratings

02

High-Value Discount Users

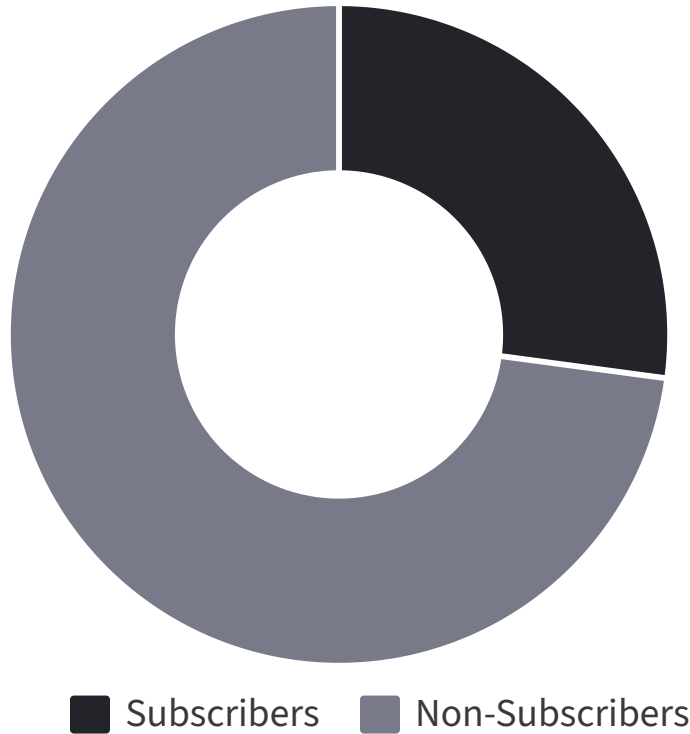
839 customers used discounts yet spent above average

04

Shipping Preferences

Express (\$60.48) vs Standard (\$58.46) average spend

Subscription Impact Analysis



The Numbers

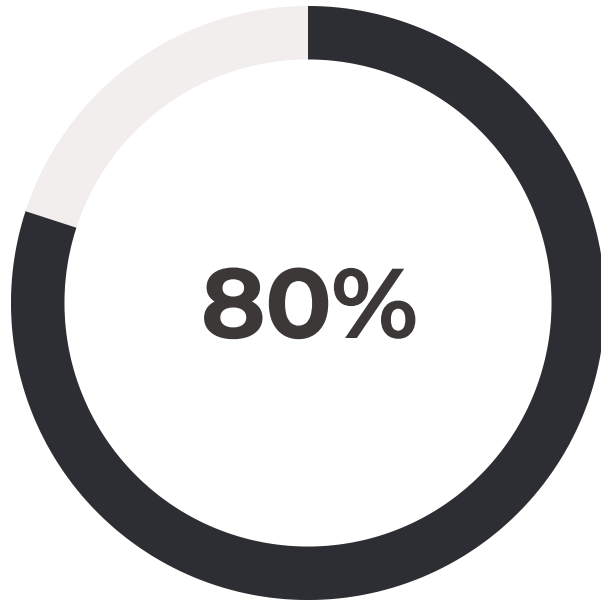
Subscribers: \$59.49 avg spend, \$62,645 total revenue

Non-subscribers: \$59.87 avg spend, \$170,436 total revenue

Insight: Similar spending patterns, but massive untapped subscription potential

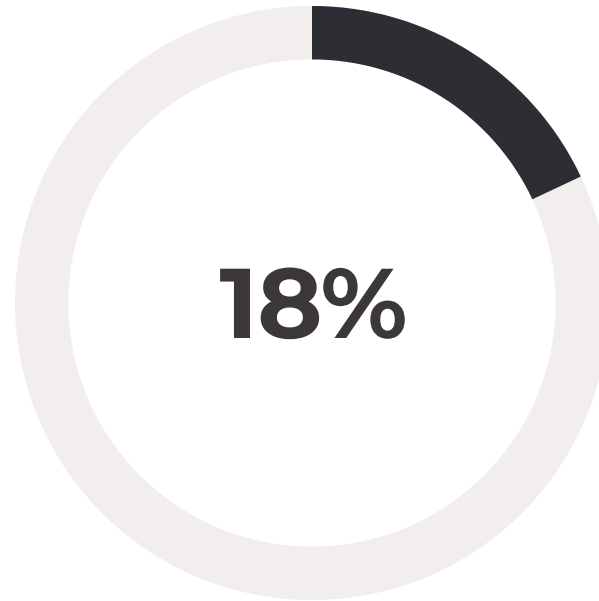
Customer Segmentation

Three Distinct Segments



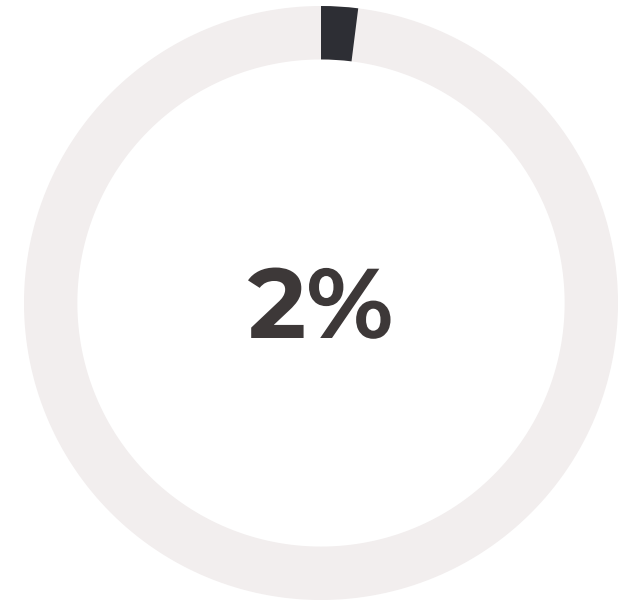
Loyal

3,116 customers



Returning

701 customers

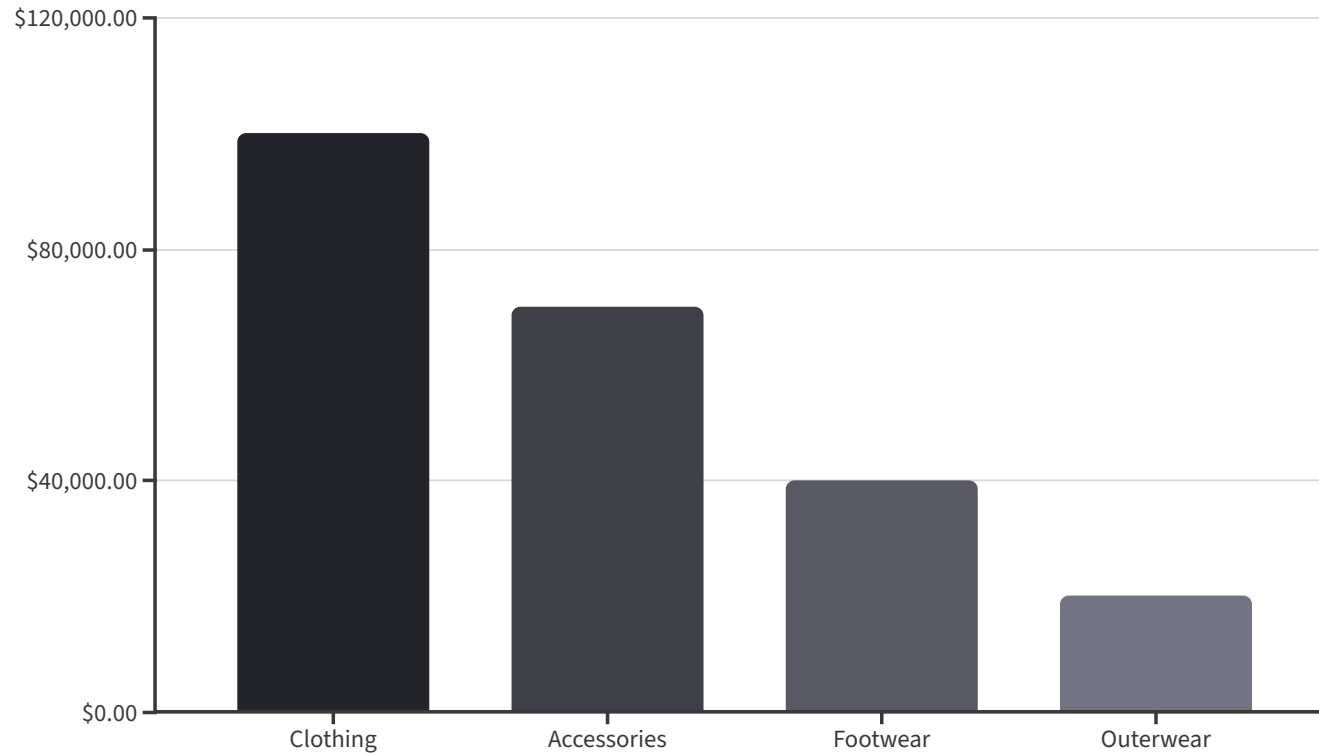


New

83 customers

Repeat buyers (>5 purchases): 958 subscribers vs 2,518 non-subscribers

Revenue Distribution Insights



Category Performance

Clothing dominates with 43% of total revenue, followed by Accessories at 30%

Age Group Leaders

- Young Adult: \$62,143
- Middle-aged: \$59,197
- Adult: \$55,978
- Senior: \$55,763

Strategic Action Plan

1

Boost Subscriptions

Promote exclusive benefits to convert 73% non-subscribers

2

Loyalty Programs

Reward repeat buyers to grow the 80% loyal segment

3

Optimize Discounts

Balance sales boosts with margin control for 839 high-value users

4

Product Positioning

Highlight top-rated items (Gloves, Sandals, Boots) in campaigns

5

Targeted Marketing

Focus on Young Adults and express-shipping customers